Foundation Sustainment Services (FS2)
Contract HM047621D0009
Aero Task Order PWS
Web Digital Vertical Obstruction File (WebDVOF) Project Addendum

21 September 2022

Version 9

Table of Contents

1 Purpose

The purpose of this Addendum is to define the expectations and requirements needed to provide Program Management, Software Maintenance, and Operations Support services as defined in the FS2 Base IDIQ and Aeronautical Task Order, for the Web Digital Vertical Obstruction File (WebDVOF) project. This Addendum establishes the acquisition framework for delivering software maintenance services required for the NGA Source Operations & Management Directorate's Foundation GEOINT Group (SF), Aeronautical Safety Office (SFA) and its mission partners.

The main objective is for the Contractor to design and execute an agile support construct to provide the optimal WebDVOF refactored cloud-optimized solution delivering user and system performance at a level equal to or exceeding the existing solution, while addressing known and upcoming security, sustainability, and compatibility issues in the legacy code base. The capabilities currently delivered by WebDVOF are to be continued in the refactored solution unless directed by the government to be deprecated. This objective covers the WebDVOF instances on all three security domains – SCI, Collateral and Unclassified.

The Contractor shall design and deliver a refactored WebDVOF solution to meet Foundation GEOINT (FG), Aeronautical Safety Office (SFA) production, research and NSG data dissemination requirements.

2 Background

The current Digital Vertical Obstruction File (DVOF) is a database of all reported vertical obstructions worldwide. Its features are divided into two capability blocks:

- A thin-client, thick-client, and machine-to-machine-accessible, web-based dissemination system for vertical obstructions (VOs) with the capability to query the dissemination database for the VOs desired, then download them in a number of userselectable formats. This site hosts the vertical obstruction data in various formats as pre-packaged downloads. It also allows customers to query and download specific areas of interest and output in a desired format.
- A thick or thin-client-accessible production system for NGA/SFA to produce vertical obstruction content that is subsequently disseminated through the WebDVOF and other NSG systems.

WebDVOF has production instances on all three NGA security domains. The provisioning environments are currently data center based, but differ across the domains. Those environments are dated and have known long-term sustainability issues that must be remediated in the near term. It has been assessed that the code base, as-is, is not viable for this, and will require significant refactoring to meet continuing mission requirements.

The end-state goal is to be accessible and usable from the future Zero-trust thin clients that will be the NGA baseline. The production and dissemination infrastructure of WebDVOF is therefore likely to be server-based and accessed through the Zero-trust thin clients. The development contractor will analyze options for the architecture and provisioning of the WebDVOF infrastructure, recommend the best approach, and implement the one selected by the government. Additionally, WebDVOF must be accessible and consumable by external entities with machine-to-machine connectivity, such as flight management systems, the Vector-Carto Production Systems (VCPS), and international partners as examples.

3 Scope

This Addendum provides the services needed for the modernization of WebDVOF. This includes but is not limited to:

1. Corrective and adaptive maintenance of the WebDVOF code baseline from the legacy, dated programming approaches such as Adobe ColdFusion and outdated operating systems, to current solution technologies and modern coding languages. This is primarily to remedy the known supportability, compatibility, performance, and security issues.
2. Adaptive maintenance to migrate that refactored code baseline from datacenter provisioning to NGA cloud provisioning.
3. Provide the necessary software install packaging of the solution stack, in a way that is compatible with the chosen operating systems and integrates into the FS2 deployment and operations baseline workflows.
4. Provide test support to establish the software test and development environment for WebDVOF.
5. Utilize the NGA CORE DevSecOps tools to the greatest degree practicable and advantageous to streamline long-term operations and sustainment.
6. Provide any other required software installation and configuration packaging for the modernized solution to enable deployment via the FS2 operations & sustainment pipeline.
7. Resolve or remove all outstanding security issues, and to perform all actions required to certify that the modernized solution meets or exceeds all NGA's security requirements.
8. Migrate the refactored WebDVOF solution to the NGA cloud environment, and provide initial provisioning of the solution stack, or assist in such as directed by the government.
9. Provide the initial Operations and Sustainment (O&S) support services for the refactored WebDVOF solution to progress to Final Operational Capability (FOC).
10. Provide legacy retirement services to decommission / remove / retire the deprecated WebDVOF solution architecture as directed by the government.
11. Provide data loading services to the Aeronautical mission area (not related to WebDVOF modernization).

The FS2 Contractor shall provide Base IDIQ SOW and Aeronautical Task Order PWS WBS 1.0, Program Management Support, WBS 4.0, Software Maintenance Services, and WBS 5.0 Operations Support Services for WebDVOF Modernization in UC, SC, TC cloud environments and meet or exceed all the requirements outlined in this Addendum, and the Aeronautical Task Order PWS.

4 WebDVOF Current State

The WebDVOF system is currently FOC on all three domains, albeit with liens against the instances for a variety of reasons. These liens are expected to continue to be allowed to be waivered while the refactored solution is in development. Reference materials for this are found at WebDVOF Upgrade - All Documents (nga.mil). The unclassified instance is at NGA: WebDVOF (Unclassified). See also Appendix E and WebDVOF Modernization - Enterprise Release Management - Confluence (nga.mil).

5 Contract Requirements and Constraints

The Contractor shall establish a software sprint maintenance cycle of a length as jointly agreed to by the Government and the Contractor. This length will be based on demonstrating satisfactory progress on remediating the outstanding liens against the legacy WebDVOF system.

At the end of each software sprint maintenance cycle, the Government and Contractor will evaluate the refactoring progress for the ability to meet mission requirements, remediate the liens, and provide cost effectiveness. If the results do not demonstrate satisfactory progress, the Government and Contractor can assess and change the software maintenance path to meet mission requirements.

5.1 Contract Requirements

The Contractor shall comply with all the Base IDIQ SOW, the IPF Capability Delivery SOP, IPF Software Solution Stack Standards, NGA Software Way, the NGA Technology Strategy, NGA Architecture Design Elements (ADE), and the NGA Web Property Standards and Guidelines (NGAI 8955.6).

The Contractor shall comply, adhere, and execute all requirements and services outlined in this Addendum and the Aeronautical Safety of Navigation Task Order PWS as follow:

a. The requirements outlined in this Addendum supersede the same requirements in the Aeronautical Safety of Navigation Task Order PWS. The definitions / descriptions for said services are superseded by this Addendum if they are modified per the tables in Section 6.

b. The Contractor is not required to deliver the other services that are in the Aeronautical Safety of Navigation Task Order PWS that are not specifically called for in this Addendum. The definitions / descriptions for the services specified in this Addendum are to be interpreted as the same as is in the Aeronautical Safety of Navigation Task Order PWS if they are not modified per the preceding condition (a).

The Contractor shall follow all the appendices in the Aeronautical Safety of Navigation Task Order PWS unless there is a specific instruction in this Addendum about the use of a specific Appendix. Additionally, the contractor shall follow the appendices in this addendum where they replace or extend those of the Aeronautical Safety of Navigation Task Order PWS.

5.2 Constraints

1) The contractor shall adhere the following governance framework:

The Contractor shall accomplish the following Phases A-B in Integration Services and must obtain permission from PMO prior to starting Phase C.

Development Phases: The modernization of WebDVOF will consist of five (5) phases that provide milestone review opportunities to the government to assess progress and satisfactory resolution to known legacy system deficiencies. These phases are:

Phase A – Conceptual Design Phase: Phase A will begin with an Initial Addendum Kickoff with the Sustainment Services PMO and the Mission Owner within 14 days of the start of this Addendum's period of performance. (CDRL A025)

During this phase, the contractor shall conduct an Analysis of Alternatives for the preferred target solution architecture, covering a minimum:

- Functional requirements baseline and identification of challenges, risks and opportunities therein
- Non-functional requirements and identification of challenges, risks and opportunities therein
- Refactoring methodologies and approaches alternatives, such as choice of programming languages, APIs, and COTS packages as examples
- Conceptual solution architecture alternatives and proposed best option

This phase will consider the known backlog of problems in the existing system, the WebDVOF capabilities and mission requirements, such trade studies as needed on topics such as provisioning, security issue remediation, market survey of COTS options over custom coding, etc. This phase will culminate in a Mission Design Review (MDR) wherein the government will review the trades and architecture alternatives results, and select the preferred target solution architecture from the options presented by the contractor, and authorize the contractor to proceed to the next phase. This phase shall cover the first 75 calendar days after the Initial Addendum Kickoff unless the PMO approves an extension. (CDRL A032).

Phase B – Preliminary Design Phase: The contractor shall begin work on refactoring / modernizing the WebDVOF solution. This phase will cover the first 120 calendar days or the end of the sprint cycle nearest to this, after the MDR decision is made. This phase will culminate in a Preliminary Design Review (PDR) wherein the government will assess the progress of the contractor in five (5) areas, and authorize the contractor to proceed to the next phase: (CDRL A032).

- The actual coding work of refactoring / modernizing the WebDVOF solution.

  ○ Demonstrate the refactoring project's JIRA action, backlog and tasking entries mapped to the target architecture agreed to by the PMO at the Phase A MDR, or updated architecture that was approved by the PMO at the relevant point during Phase B. ○ Example code and initial execution demonstration of what has been accomplished to date. ○ Demonstrate the code and software baseline configuration management approach (e.g. the GITLAB project with source code, etc.).
- Completeness of mission capabilities and requirements within the portions of WebDVOF refactored / modernized at that point.
- Implementation risks and problems encountered, and the resolution thereof.
- Progress vs the project schedule.
- Progress vs incurred cost to date.

<u>The Contractor shall accomplish the following Phases C-E in Operations Support. The contractor shall obtain permission from PMO prior to executing Phases C-E.</u>

<u>Phase C – Critical Design Phase</u>: Subject to favorable results of the PDR review, the contractor shall continue work. This phase is the remaining part of the WebDVOF modernization work, covering 150 days after PDR unless the PMO approves an extension. During this phase, the contractor shall submit progress updates to the PDR package, aligned to the five evaluation areas from Phase B, at the conclusion of each major sprint cycle in the development work. These updates will not be formal review events, however, the government may elect to make any of them such if the results at that point indicate a need to examine progress in a more formal sense. (CDRL A032).

<u>Phase D – Final Demonstration and Test Phase</u>: Upon completion of the WebDVOF modernization work, the full WebDVOF end-to-end capability shall be demonstrated to the government. It shall also be evaluated by the mission users (e.g. SF, DoD customers, commercial/civil customers, international partners for example) in a formal test that will culminate in full system IOC acceptance by the SF mission partner. The length of this phase will be determined by customer testing and certification needs. (CDRL A032).

<u>Phase E – Transition to Operations and Sustainment</u>: Upon full system IOC acceptance by the SF mission partner, the WebDVOF modernization effort will transition the solution to the long-term Operations and Sustainment team. This long-term work will be done under a separate addendum. At transition + 90 calendar days unless the PMO

approves an extension, full system FOC will be declared if deemed acceptable by the SF mission partner. The FOC event will end the WebDVOF modernization effort. (CDRL A032).

2) This effort will cover all three security domains for WebDVOF. The order in which the contractor will prioritize the domains will be set by the PMO as part of the Phase A MDR decision.
3) The vendor shall use the NGA CORE capabilities, services and environments in preference to self-provisioning unless compelling business case is made to the alternative, and approved by the government.
4) The solution / process shall comply with the standards and references per Section 7

6 Services

6.1 Program Management Support (WBS 1.0)

The Contractor shall provide Program Management Services (WBS 1.0) for the modernization effort per the Aeronautical Safety of Navigation Task Order PWS, section 6.1. (CDRL A001)

6.2 Software Maintenance Services (WBS 4.0)

The Contractor shall provide the Software Maintenance services (WBS 4.0) for WebDVOF instances in the unclassified, collateral and SCI environments per Section 6.2 of the Aeronautical Safety of Navigation Task Order PWS. (CDRL A009, A010, A011, A012, A014, A022, A044)

Furthermore, the specifications of the services therein are extended to additionally include the following:

FS2 WBS#	WBS Title	Software Maintenance Service Descriptions
4.12	Demo Support (A041, A044)	The contractor shall provide Demo Support as required by the government to show proposed mitigation approaches to resolve the liens and demonstrate progress against the same. The contractor shall provide such interim demonstrations as requested by the government during Phases A-C per section 5.2. The contractor shall provide an end-to-end demonstration of the full modernized WebDVOF solution at Phase D per section 5.2.
4.13	Test Support (A027, A041, A044)	The contractor shall provide a final end-to-end test and evaluation event at Phase D per section 5.2.

FS2 WBS#	WBS Title	Software Maintenance Service Descriptions
4.16	Corrective Maintenance (A004, A005, A009, A011, A022, A033, A034, A039)	The contractor shall ensure the refactored solution addresses the known liens requiring corrective actions. The contractor need not identify the root cause of the lien in the legacy system code base if the applicable code section therein is not being reused in part or whole.
4.17	Adaptive Maintenance (A004, A005, A009, A011, A022, A033, A034, A039)	The contractor shall ensure the refactored solution is coded using modern programming languages and is compatible with cloud provisioning environments. The contractor shall follow the governance framework as described in Section 5.2, providing the evidences at each milestone event as described therein.
4.18	Preventive Maintenance (A004, A005, A009, A011, A022, A033, A034, A039)	The contractor shall examine the refactored solution approach and identify any dependencies or third-party usages that either currently have supportability, sustainability, or security issues, or are likely to have such in the near future. At the direction of the government, the contractor shall identify alternative third party service providers, or develop embedded WebDVOF solutions to provide the capability.
4.19	Perfective Maintenance (A004, A005, A009, A011, A022, A033, A034, A039)	The contractor shall evaluate the architecture and structure of the legacy WebDVOF solution for efficiency and effectivity, and recommend changes to it where applicable for the refactored solution that perfects the WebDVOF solution architecture. The contractor will implement said changes in the refactored solution upon approval by the government.

4.36	Requirements Analysis (A009, A011, A022, A033)	The contractor shall evaluate the legacy WebDVOF solution capabilities for efficiency, effectivity, suitability and applicability to the WebDVOF mission (for example, identify if capabilities are in the WebDVOF baseline that are no longer used by the SF mission customer, or could be deprecated if the refactored solution were to be structured in a particular way). The contractor will implement said changes in the refactored solution upon approval by the government The contractor shall conduct such analysis and trade studies as required to meet the Phase A / MDR governance needs as described in Section 5.2.

Table 6-1: Software Maintenance requirements.

6.3 Operations Support Services (WBS 5.0)

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The Contractor shall provide the Software Maintenance services (WBS 5.0) for WebDVOF instances in the unclassified, collateral and SCI environments per Section 6.3 of the Aeronautical Safety of Navigation Task Order PWS.

Furthermore, the specifications of the services therein are extended or amended to include the following:

FS2 WBS#	WBS Title	Operations Support Service Descriptions
5.1	System Performance Optimization (A037, A044)	In particular, the contractor shall evaluate the refactored solution performance against the legacy solution performance to ensure at-least-equal, and preferably increased, system performance.
5.4	Automation (A009, A039)	As part of Phase A, the contractor shall evaluate the use of the NGA CORE capabilities for suitability to use in the deployment and operations of the modernized WebDVOF solution, including the use of the common provisioning environments.
5.11	DevSecOps Management (A009, A039)	As part of Phase A, the contractor shall evaluate the use of the NGA CORE capabilities for suitability to use in the deployment and operations of the modernized WebDVOF solution, including the use of the common provisioning environments.
5.20	Help Desk and Desk Side Support: Tier I Support (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.
5.21	Help Desk and Desk Side Support: Tier 2 Support (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.

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5.22	Help Desk and Desk Side Support: Tier 3 Support (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.
5.23	Incident Tracking (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.

FS2 WBS#	WBS Title	Operations Support Service Descriptions
5.24	Discrepancy Report (DR) Investigation (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.
5.25	System Performance (A030, A037)	In particular, the contractor shall evaluate the refactored solution performance against the legacy solution performance to ensure at-least-equal, and preferably increased, system performance.
5.26	Service Status/ Status Tracking (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.
5.27	Metric Collection and Analysis (A030, A037)	The contractor shall also report metrics according to the assessment areas for the milestones as defined in Section 5.2.
5.28	Cyber Security (A005, A019, A046)	The contractor shall particularly focus on the known deficiencies in the legacy system to ensure they are no longer relevant / removed from / remediated from the modernized WebDVOF solution.

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FS2 WBS#	WBS Title	Operations Support Service Descriptions
5.29	Authorized Outage Coordination (A030)	The contractor shall provide this service only for the modernized WebDVOF instance, during Phase E in the 90-day period leading to FOC.
5.37	Cloud Engineering, Integration, and Management (A006, A009, A019, A021)	The contractor shall also produce the target architecture analysis of alternatives per the definition of Phase A in Section 5.2.
5.38	Engineering (A019, A046)	The contractor shall also produce the target architecture analysis of alternatives to remediate known software, programming language, and security issues in the legacy system per the definition of Phase A in Section 5.2.
5.39	Technical Writing/Documentation Support (A032)	The contractor shall also provide the milestone review artifacts per the five Phases as defined in Section 5.2.
5.41	Transition-In (A006, A041)	The contractor shall provide this service in the period leading up to, and during, Phase E to transition the modernized WebDVOF solution to the long-term sustainment and operations team.

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5.42	Transition-Out (A012)	The contractor shall provide this service in the period leading up to, and during, Phase E to transition the modernized WebDVOF solution to the long-term sustainment and operations team, if directed to do so by the government, and to the extent that it assists in promoting the modernized WebDVOF solution to operations. Removal and retirement of the legacy WebDVOF solution will mostly be the responsibility of the long-term operations and sustainment team such as it is the responsibility of FS2. Where the legacy equipment is not the responsibility of FS2, the contractor will provide such support to a third-party contractor to the degree directed by the government.

Table 6-2: Operations Support requirements.

6.4 Operations Support Services (WBS 5.0)

The contractor shall provide support of Aeronautical Safety of Navigation data loading capability per Section 6.3 of the Aeronautical Safety of Navigation Task Order PWS. This capability follows a workflow as defined in Appendix I, but may be modified based on the modernization of the WebDVOF per this Addendum, or other Aeronautical-related systems directly affecting this capability.

Furthermore, the specifications of the services therein are extended or amended to include the following:

FS2 WBS#	WBS Title	Operations Support Service Descriptions
5.25	System Performance (A037)	The contractor shall provide System Performance services in accordance with FS2 Base IDIQ SOW section 6.25 to

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		perform ingest of Aeronautical data as defined in this section and in Appendix I. The contractor shall operate and maintain the Aero SoN capabilities and services at a user operational availability rate of 99.9% for deployed services. The contractor shall provide the Performance Metrics as they are listed in Appendix A of this task order.
5.33	Database Management Services (A037)	The contractor shall provide Database Management Services for the Aero SoN operational capabilities in accordance with FS2 Base IDIQ SOW section 6.33 for the ingest and management of Aeronautical data as defined in this section and in Appendix I. The contractor shall assume responsibilities for creating, configuring, administering, and maintaining the enterprise baseline mission domain geodatabases used by viewers, editors, and analysts.

7 List of WebDVOF Applicable Documents

The Table is Unclassified

No	Document Title
1	GENERAL SPECIFICATION FOR VECTOR PRODUCT FORMAT (VPF) PRODUCTS, MILPRF-0089049
2	PERFORMANCE SPECIFICATION SHEET VECTOR VERTICAL OBSTRUCTION DATA (VVOD), MIL-PRF-89049/9A
3	NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY PRODUCT SPECIFICATIONS FOR VERTICAL OBSTRUCTIONS (AERONAUTICAL), 2nd EDITION

Table 7-1- WebDVOF Reference Documents

UNCLASSIFIED

8 List of CDRLs

This addendum uses the CDRL list from the Base IDIQ and does not have any extra CDRLs unique to this addendum. CDRLs are noted in each section of the WBS

CDRL Title	CDRL #	Data Item Description	Reference	Initial Addendum Award	Subsequent Versions
End of Sprint Status Reports	A004	CFA	6.2	n/a	End of Sprint
Security Plan	A005	CFA	6.2, 6.3	Draft at Start of Phase B	End of each Sprint in which cyber / information security is affected
Transition-In Plan	A006	CFA	6.3	Draft at Start of Phase C	Final by Start of Phase D
Cloud Migration Plan	A009	CFA	6.2, 6.3	Award +60 Calendar Days	As directed by the Government
Software Code and Software Code Documentation	A011	CFA	6.2	Award +60 Calendar Days	End of each Sprint
Transition-Out Plan	A012	CFA	6.3	Draft at Start of Phase C	Final by Start of Phase D
Version Description Document	A019	CFA	6.3	Award +60 Calendar Days	End of each Phase
Release Plan	A021	Update JIRA and a PDF copy to PMO	6.3	5 calendar days after first RPE	5 calendar days after each RPE
Value Engineering Change Proposal	A022	CFA	6.2	n/a	As appropriate during Phase A-E

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Task Order Kickoff Presentation	A025	Power Point	5.2	At the Kickoff meeting	15 calendar days after future subsystem is added to the Task-Order
Test Readiness Review checklist	A027	CFA	6.2	Deliver to the Government 30 calendar days before the first TRR event	Deliver to the Government 30 calendar days before each TRR event
Operations Support Plan (OSP)	A030	CFA	6.3	Draft at start of Phase C	Final by Start of Phase D
Design Review Presentation (DRP)	A032	CFA	5.2, 6.3	At each Phase Review Milestone	After Phase Review Milestone updated as directed by the PMO
Requirements Traceability Matrix (RTM)	A033	CFA	6.2	At end of Phase A	Updated at end of each Sprint as needed
Capability Integration Schedule (CIS)	A034	CFA	6.2	At end of Phase A	Updated at end of each sprint thereafter
System Performance Optimization Technical Report	A037	CFA	6.3, 6.4	n/a	As directed by the Government
Automation Deployment Instructions	A039	CFA	6.2, 6.3	First Deployment	Every deployment
Operational Readiness Review (ORR) Checklist	A041	CFA	6.2, 6.3	15 calendar days prior to readiness event	Updated after test event as directed by PMO
Integration Test Report	A044	CFA	6.2, 6.3	As directed by Government	As directed by the Government

UNCLASSIFIED

Corrective Action Report (CAR)	A046	CFA	6.3	As directed by Government	As directed by the Government

9 List of Appendixes

This table is Unclassified

Appendix	Documents
List of Appendixes in the WebDVOF Addendum	
Appendix E – WebDVOF Unclassified Architecture	WebDVOF Addendum
Appendix F - Performance Metrics	WebDVOF Addendum
Appendix G - WebDVOF Quality Assurance	WebDVOF Addendum
Appendix H – WebDVOF Existing Functionality	WebDVOF Addendum
Appendix I – Data Loader Workflow Description	WebDVOF Addendum
List of Appendixes in Aeronautical Safety of Navigation Task Order PWS	
Appendix A: Performance Metrics	Aeronautical Safety of Navigation Task Order
Appendix B: CDRLs	Aeronautical Safety of Navigation Task Order
Appendix C: Technical Library Contents List	Aeronautical Safety of Navigation Task Order
Appendix D: Key Personnel Job Descriptions and Qualifications	Aeronautical Safety of Navigation Task Order

Table 9-1 – List of Appendixes

Appendix E – WebDVOF Unclassified Architecture

WebDVOF DMZ Implementation (draft, version 13, 2022-04-20c)

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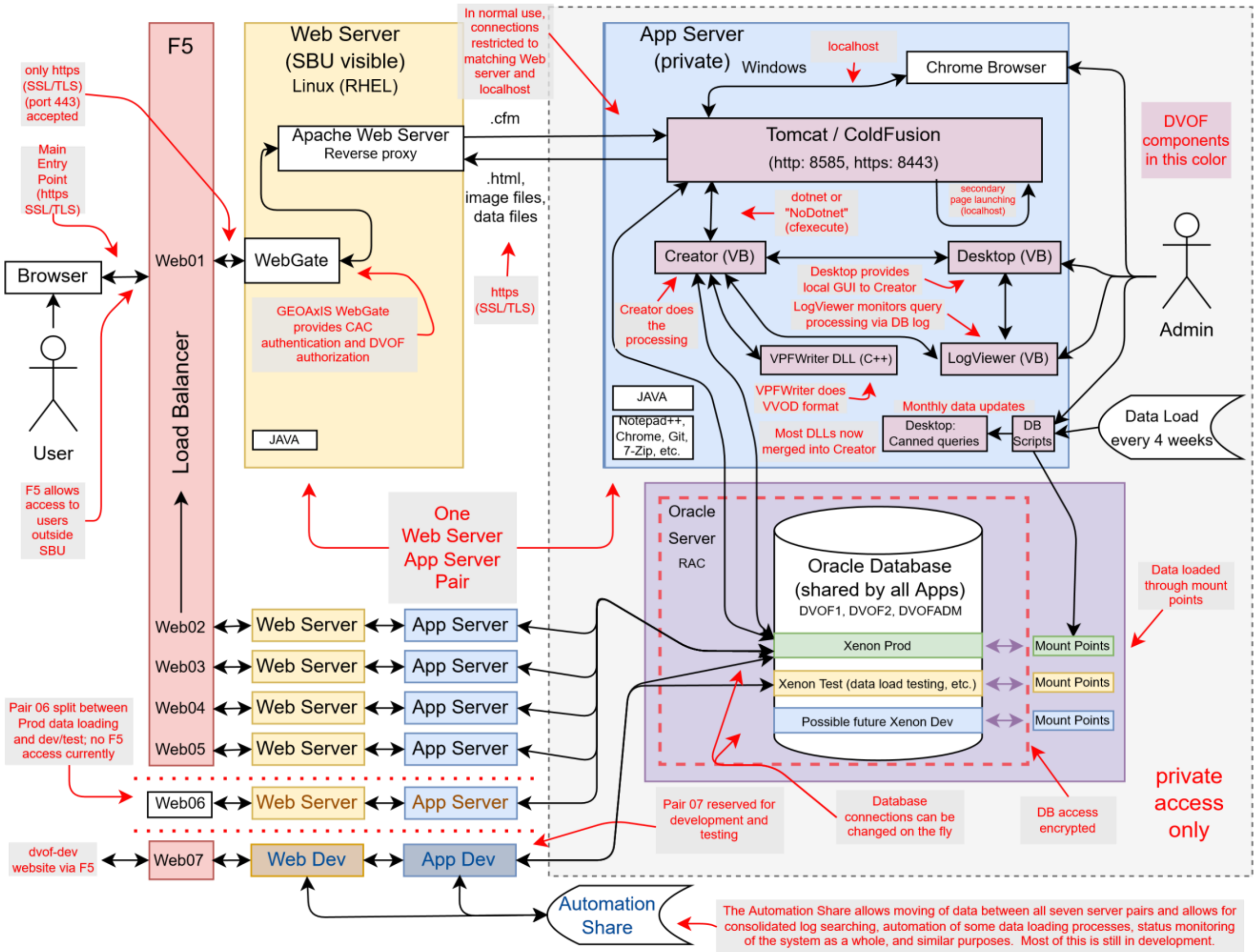

Notes:

1. **Flow:** (1) User navigates to WebDVOF URL and submits a request to Web Server proxy thru F5 load balancer. (2) Apache Reverse Proxy authenticates and authorizes via WebGate and GEOAxIS. (3) Apache Proxy forwards to Tomcat and ColdFusion for parsing. (4) ColdFusion calls VB app Creator to process queries. (5) Output pages and links to output data passed back to Apache, then passed to user.

2. **Server Layout:** One-for-one mapping of Web Server to App Server. This allows proper mapping of a user request with the server pair doing the processing. As processing is stateful, i.e. output products are only available on that server pair, user must remain on same pair for entire session. DotNet forced single threading. Now using "NoDotNet" enhancement, which does allow a server pair to process multiple user requests at same time.

3. **Access Control:** Access to system is limited by WebGate (plug-in to Apache on Web Server) to those on WebDVOF user list. CAC required for access. F5 now provides access to users outside SBU. Additional access restrictions for pair 07 to be implemented at Tomcat, limiting App07 access to just developers.

4. **DOTNET and NoDotNet:** The Creator app is a Visual Basic Assembly and, in the original design, interacts with ColdFusion via the ColdFusion DOTNET service. Most DLLs are gone, merged into Creator, in

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part because the DOTNET service can't handle DLLs passing complex parameters. Issues with DOTNET becoming unresponsive required the DOTNET service to be restarted before each run, limiting an App server to processing one run at a time; i.e. multiple users can't run queries on the same App server at the same time. In response, a "NoDotNet" enhancement was designed and implemented, removing most uses of DOTNET. Most synchronization between ColdFusion and Creator is now handled by a set of control files and handshakes. This is now the standard configuration.

5. **Admin Access:** Expect all admin tasks, such as monthly data loading, to be done locally on the App servers. This requires admins to be able to access the Xenon App servers through Shield. Due to limitations of Shield, alternatives, such as service accounts, are now being looked at to, for instance, do the data load processing.

6. **See also** WebDVOF Modernization - Enterprise Release Management - Confluence (nga.mil) and NGA: WebDVOF (Unclassified)

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Appendix F - Performance Metrics

The Contractor shall meet or exceed all the Performance Metrics in this table and the Performance Metrics in the Aero Task-Order PWS (where applicable), except for the Performance Metrics related to the Integration Services (WBS #03).

For WebDVOF specifically, see WebDVOF Performance - Enterprise Release Management - Confluence (nga.mil)

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Appendix G - WebDVOF Quality Assurance

This Addendum will use the Aeronautical Task Order QASP.

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Appendix H – WebDVOF Existing Functionality

See WebDVOF Modernization - Enterprise Release Management - Confluence (nga.mil) and NGA: WebDVOF (Unclassified)

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Appendix I – Data Loader Workflow Description

The Data Loader task has a workflow as follows:

- Load aeronautical products received on transfer drive, FTP sites, CDs and DVDs from multiple sources to the NGA file servers and operational data stores via workstation on all three (3) NGA security domains. Transfer/transmit data and products to other Aeronautical and NGA data stores/dissemination sites (ie., MOW, ACES). Products are received weekly, every 28 days in accordance with the Aeronautical Information Regulation and Control (AIRAC) cycle and on an ad-hoc basis. o Data loaders shall implement programming, as required, to troubleshoot, identify and resolve issues with load scripts or data that may cause loads to fail.
  - Data loaders shall load products to the UC2S, SC2S and TC2S cloud environments as they become available and shall document load processes.

- Load database dumps/files received on transfer drive, CDs and DVDs from multiple sources via their workstation into Oracle and/or PostGIS databases on all three (3) NGA security domains. Data loaders shall have some database/SQL background as they shall be required to troubleshoot, identify and resolve issues with load scripts or database dumps/files that may cause loads to fail. Database dumps/files are received weekly, every 28 days in accordance with the Aeronautical Information Regulation and Control (AIRAC) cycle and ad-hoc occurrences.

- WebDVOF (Digital Vertical Obstruction File) is a web application providing aviators and mission planners access to a world-wide database of vertical obstruction data and custom generated downloads. WebDVOF stores, retrieves, and distributes data concerning known vertical obstructions worldwide. Create outputs for the following products in accordance with the Aeronautical Information Regulation and Control (AIRAC) 28-day cycle or weekly as required as part of the data load process: o Digital Vertical Obstruction File (DVOF), o Vertical Vector Obstruction Data (VVOD), o Obstruction Change File (OCF),
  - Tab Formatted Aeronautical Dataset –Obstructions (TFADS-O) o Shapefiles (Lines and Points)

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Program	POCs	Product(s)	SFA Websites **	Production timeline	Must be loaded...	Notes/Load requirements
Airfields	Megan Weymouth Nick Jones	AAFIF AFD	ACES 2.0 Aerospatial* CARDS*	AAFIF - every 7 days AQP - every 28 days AFD - as required	Upon receipt	
Digital Aeronautical Flight Information File (DAFIF)	David Atkinson Rick Timms	DAFIF 8.0 DAFIF 8.1 DAFIF 9 (summer 2022)	ACES 2.0 Aerospatial* CARDS*	Every 28 days	NLT 10 prior to effective date or upon receipt	
Dissemination	Edwin "Bruce" Farnham Alexandra Marks	Aero Mobile App - iOS Aero Mobile App - Windows Aero Mobile App - Android Aero Data Server - Mac Aero Data Server - PC	ACES 2.0 Aerospatial* CARDS*	Every 28 days	NLT 10 prior to effective date or upon receipt	
Flight Information Publications (FLIP)	Cameron Korrect	FLIP Planning FLIP Enroute Supplements FLIP Enroute IFR Charts AACDTD MTR charts FLIP DVD FAA FLIP Canadian FLIP	ACES 2.0 Aerospatial* CARDS*	Every 28-56 days	NLT 10 prior to effective date or upon receipt	
Seamless Enroute Charts	Cameron Korrect	Seamless enroute charts - various scale codes	ACES 2.0 Aerospatial* CARDS*	Every 28 days	NLT 10 prior to effective date or upon receipt	Navy Research Labs project; still in beta
Terminals	Ron Brumback	FLIP Terminals E-IPL plates E-IPL DVD	ACES 2.0 Aerospatial* CARDS*	Every 28 days	NLT 10 prior to effective date or upon receipt	
Vertical Obstructions (VO)	VO Program Officer Nick Jones	OCF VVOD TFADS-O ESRI Shapefiles Custom Media	ACES 2.0 WebDVOF Aerospatial* CARDS*	Every 28 days	NLT 10 prior to effective date or upon receipt	
CADRG/ECRG	Steve Rogan (STP2)	CADRG charts ECRG charts	ACES 2.0 Aerospatial* CARDS*	Monthly	Upon receipt	Not an SFA program or product
* These legacy websites will be integrated into ACES 2.0 and shut down by EOY CY21.						
** NIPR, SIPR, and JWICS enclaves, as appropriate or available						

- Maintain the component capability of the systems that support Aeronautical dissemination ○ GeoServer
  - Oracle Tables, and associated views, indices, triggers and stored procedures ○ PostGres/PostGIS
  - ArcGIS Server